

1 What is BIOS?

BIOS stands for

Basic Input Output System

BIOS is a **firmware** stored on a **ROM/Flash chip on the motherboard**.

◆ What BIOS does (in simple words):

When you press the **power button**:

1. BIOS gets power first
2. Checks hardware (**POST – Power On Self Test**)
3. Initializes CPU, RAM, keyboard, disk, display
4. Finds **boot device** (HDD / SSD / USB)
5. Loads the **Operating System**



Without BIOS → **Computer cannot start**

2 Types of BIOS Modes

There are **TWO main BIOS modes** used in modern computers:

◆ 1. Legacy BIOS (Traditional BIOS)

- ✓ Old system (used before 2012)
- ✓ Text-based interface
- ✓ Keyboard only
- ✓ Slower boot

Limitations:

- Supports max **2 TB disk**
- Only **4 primary partitions**
- No Secure Boot

◆ 2. UEFI BIOS (Modern BIOS) Unified Extensible Firmware Interface

- ✓ New system (used in Windows 10 / 11)
- ✓ Graphical interface
- ✓ Mouse + keyboard
- ✓ Faster boot
- ✓ Secure & advanced

Advantages:

- Supports **very large disks (up to 9.4 ZB)**
- Supports **128 partitions**
- Secure Boot support
- Faster startup

3 Important BIOS Settings (Very Important for Interviews)

◆ **Boot Settings**

- Boot Priority (HDD / SSD / USB / Network)
- Boot Mode: **Legacy / UEFI**
- Secure Boot Enable / Disable

◆ **Storage Settings**

- SATA Mode:
 - IDE (Old)
 - AHCI (Best & modern)
 - RAID

◆ **Security Settings**

- BIOS Password
- Secure Boot
- TPM (Trusted Platform Module)

◆ **Power Settings**

- AC Power Recovery
- Wake on LAN
- Fan control

◆ Hardware Settings

- CPU virtualization (VT-x / AMD-V)
- USB enable/disable
- RAM frequency

4 BIOS Mode vs Partition Type (MOST IMPORTANT QUESTION)

This is a **very common interview & practical question**.

◆ Partition Types

There are **TWO** partition styles:

Partition Type Full Form

MBR Master Boot Record

GPT GUID Partition Table

🔥 Which Partition Type Works with Which BIOS Mode?

BIOS Mode	Supported Partition Type	Used In
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Legacy BIOS	MBR only	Old systems
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UEFI BIOS	GPT only (Recommended)	Modern systems
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📌 Important rule to remember

❌ Legacy BIOS ❌ cannot boot from GPT

❌ UEFI ❌ cannot boot from MBR (Secure Boot ON)

5 Detailed Comparison (Exam Ready)

Feature	MBR	GPT
Max Disk Size	2 TB	9.4 ZB
Max Partitions	4	128
Boot Mode	Legacy	UEFI

Security	Low	High
Redundancy	No	Yes (backup headers)

6 Real-Life Examples

◆ Windows 7 Old PC

- BIOS Mode: **Legacy**
- Partition Type: **MBR**

◆ Windows 10 / 11 New Laptop

- BIOS Mode: **UEFI**
- Partition Type: **GPT**

◆ Secure Boot Enabled System

- BIOS Mode: **UEFI**
- Partition Type: **GPT only**

7 How to Check in Windows (Practical)

◆ Check BIOS Mode

msinfo32

Look for **BIOS Mode**:

- Legacy
- UEFI

◆ Check Partition Type

diskpart

list disk

✓ If disk has * under GPT column → GPT

✗ No * → MBR

8 One-Line Interview Answer (Quick Revision)

Legacy BIOS uses MBR partition and UEFI BIOS uses GPT partition.